

CODESENTINEL

Autonomous AI Code Review & Self-Debugging Audit Report | Offline Engine: Qwen2.5-Coder (1.5B)

Audit Timestamp: 2026-09-10 12:35:29

Execution Mode: 100% On-Device / Air-Gapped / Zero Cloud Leakage

Code Health Score

60 / 100

Confirmed Defects

2 Real Bug(s)

Noise Filtered

0 Discarded (0%)

Automated Patch

PASSED

Agentic Self-Debugging Pipeline Execution:



Codebase Structure & Usage Intelligence Audit:

Imported Packages & Libraries:
math, sys

Detected Functions & Signatures:
multiply(a, b); divide(a, b)

Lines of Code (LOC) Metrics:
Total Lines: 8 | Code: 6 | Comments: 1 (16.7%)

Cyclomatic Complexity & Maintainability:
Branching Score: 2 - Rating: Low (Easily Maintainable)

Visual Defect Analytics & Four Illustration Graphs:



Defect Verification Breakdown Matrix:

Stage	Evaluation & Reasoning Detail	Defect Status	Remediation
Step 1	1. Multiply function adds instead of multiplying. 2. Divide function does not check for zero division.	Potential Bug	Identified
Step 2	1. CONFIRMED: Multiply uses addition operator instead of multiplication. 2. CONFIRMED: Divide lacks a zero divisor check.	CONFIRMED	Isolated
Step 3	Synthesized patch addressing verified defects without side-effects	Auto-Patched	Implemented
Step 4	FIX LOOKS CORRECT: All arithmetic and division safeguards are properly implemented.	PASSED	Audited

Step 1: Suspected Bug Scan

- 1. Multiply function adds instead of multiplying.
- 2. Divide function does not check for zero division.

Step 2: Verification & Noise Reduction

- 1. CONFIRMED: Multiply uses addition operator instead of multiplication.
- 2. CONFIRMED: Divide lacks a zero divisor check.

Step 3: Synthesized Code Patch

```
def multiply(a, b):  
    return a * b  
  
def divide(a, b):  
    if b == 0:  
        raise ValueError('Cannot divide by zero')  
    return a / b
```

Step 4: Regression Validation & Sanity Verdict

FIX LOOKS CORRECT: All arithmetic and division safeguards are properly implemented.

Official Audit Seal: PASSED (Autonomous Multi-Step Verification)

This software artifact was audited and self-debugged using the CodeSentinel local agentic pipeline.